IPB University | Tri-U 32

TRI-U INTERNATIONAL JOINT SEMINAR & SYMPOSIUM 2026 · CHIANG MAI UNIVERSITY, THAILAND

HEALTHCARE INNOVATION

An exploratory multi-scale analysis linking MRI radiomics with KDR/VEGFR-2 expression and kinase-domain structure in glioblastoma

Muhammad Dandy Nugraha · Wisnu Ananta Kusuma ·
Department of Bioinformatics, IPB University

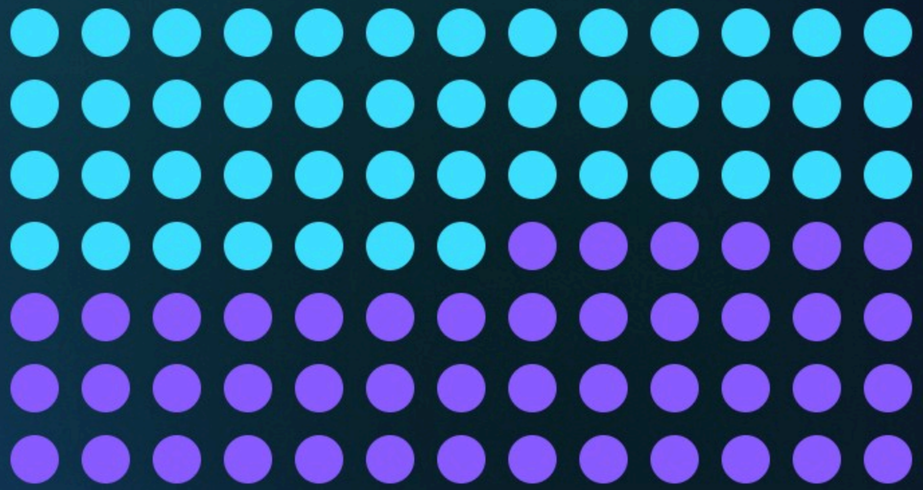
15 months

MEDIAN SURVIVAL, GLIOBLASTOMA



Background: illustrative animation, not data.

KDR EXPRESSION · MEDIAN SPLIT



46 HIGH 45 LOW

THE QUESTION

Measure KDR from an MRI, not from tissue.

Bevacizumab	Blocks the signal that grows new blood vessels into the tumour.
VEGFR-2 · KDR	The receiver, and the gene that builds it.

91 PATIENTS · ALL PUBLIC DATA

No new scans, no new procedures, nothing a hospital does not already have.

BACKGROUND

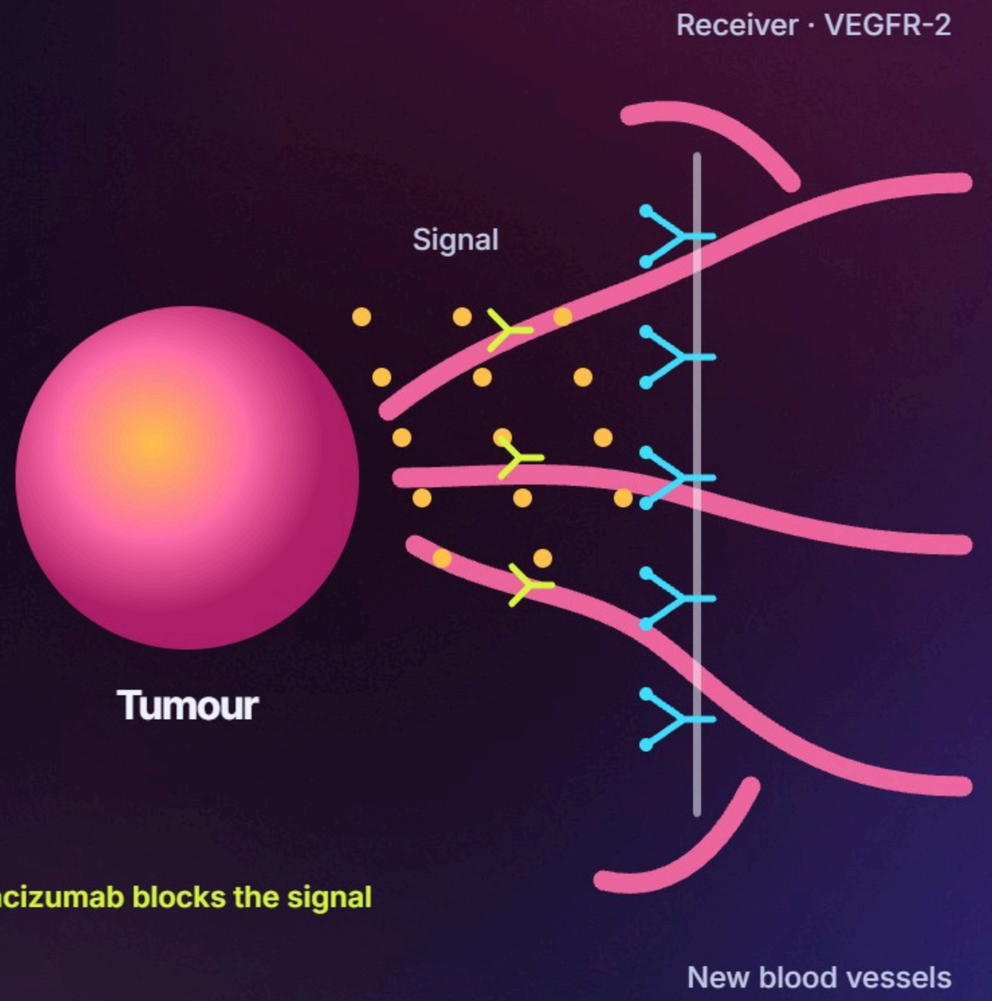
A signal does nothing unless something receives it.

01 A tumour grows faster than its own blood supply.

02 It sends a signal for new blood vessels: angiogenesis.

03 More receivers means more blood vessels, and more for the drug to block.

That is why KDR is our target, and why reading it without surgery would be useful.



METHODS

Image, gene, structure.

PyRadiomics turns each MRI into measurements we can name. Does it rise and fall together with KDR in the same patient?

Image
MRI · T1, T1-Gd, T2, FLAIR

704 pre-computed · 1,098 PyRadiomics descriptors

Gene
Spearman · Benjamini-Hochberg · q < 0.05

Random Forest · nested

Structure
PDB 2OH4 · 2.05 Å · PyMOL

> **Why not deep learning? Ninety one patients is far too few.**
>
> And a deep learning model cannot tell a doctor which part of the image it used.

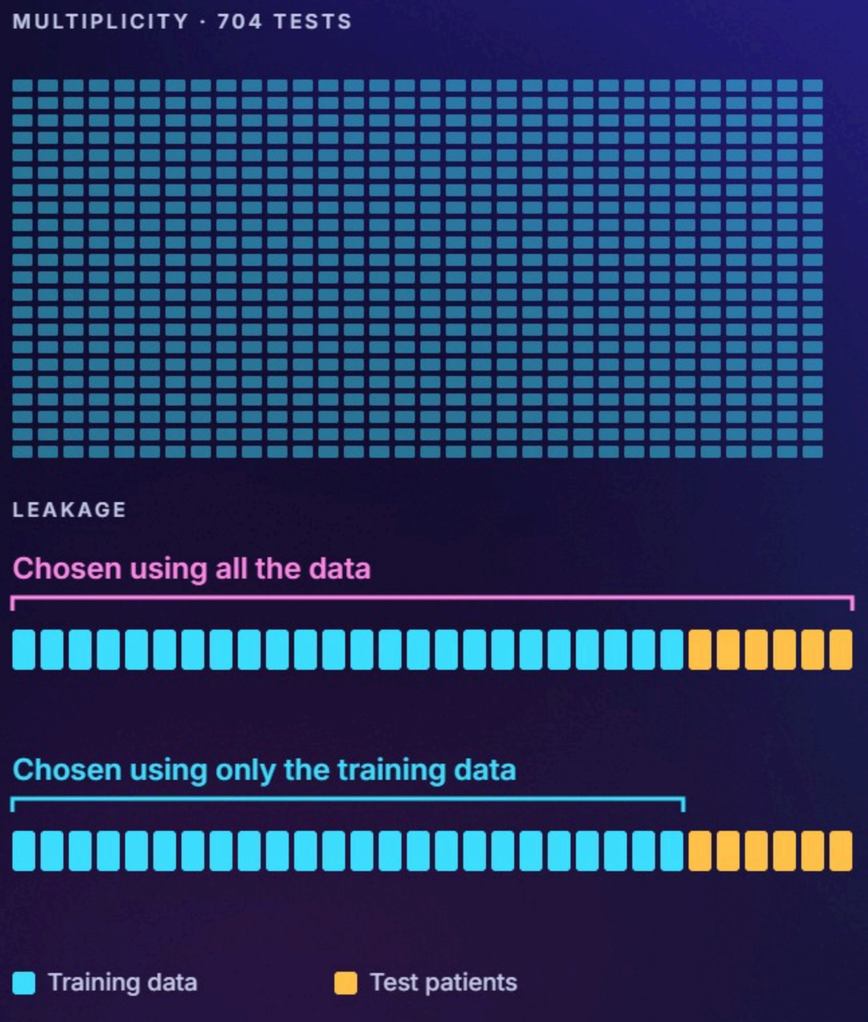
TWO CHECKS

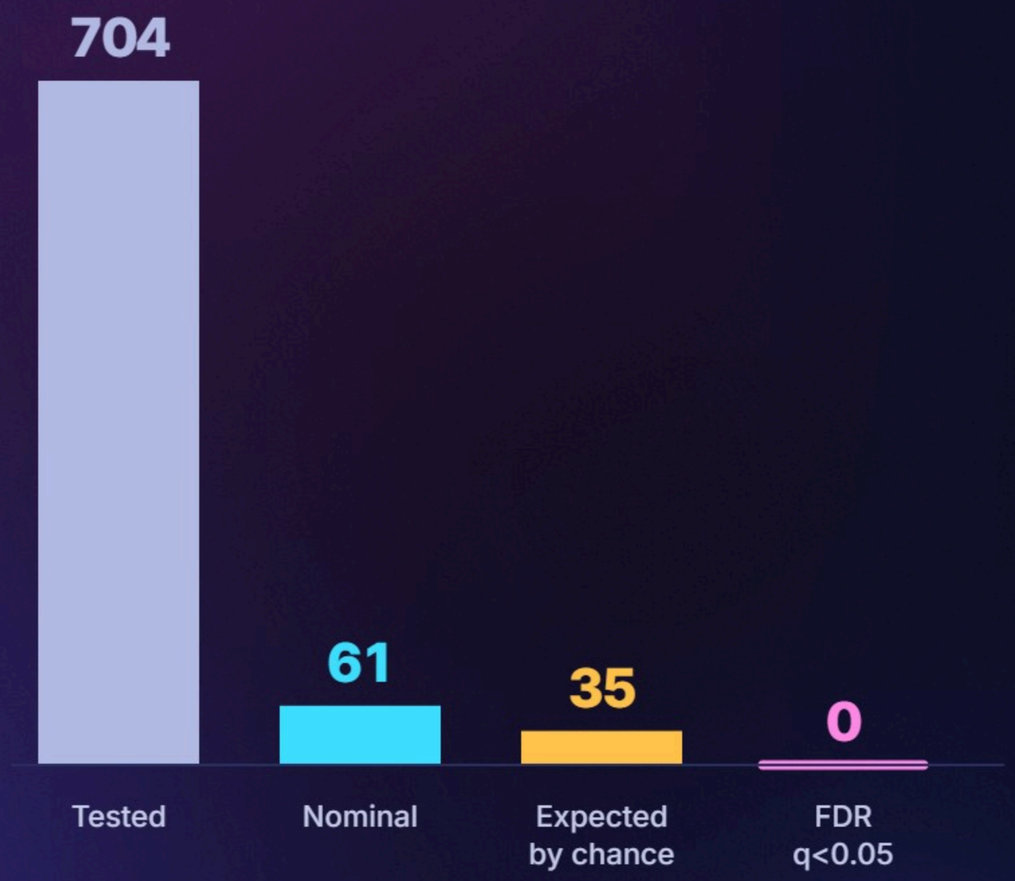
Two checks. Neither was in my first draft.

01 Multiplicity

02 Leakage

Both are standard. And together, they are the reason this talk exists.





CHECK 01 · MULTIPLICITY

Zero survived.

704	61	35	0
TESTS, ONE PER MEASUREMENT	CAME BACK SIGNIFICANT	EXPECTED BY CHANCE ALONE	SURVIVED FDR CORRECTION

If nothing were true at all, chance alone would still give us about thirty five.

Not one out of seven hundred and four. Strongest: q = 0.530

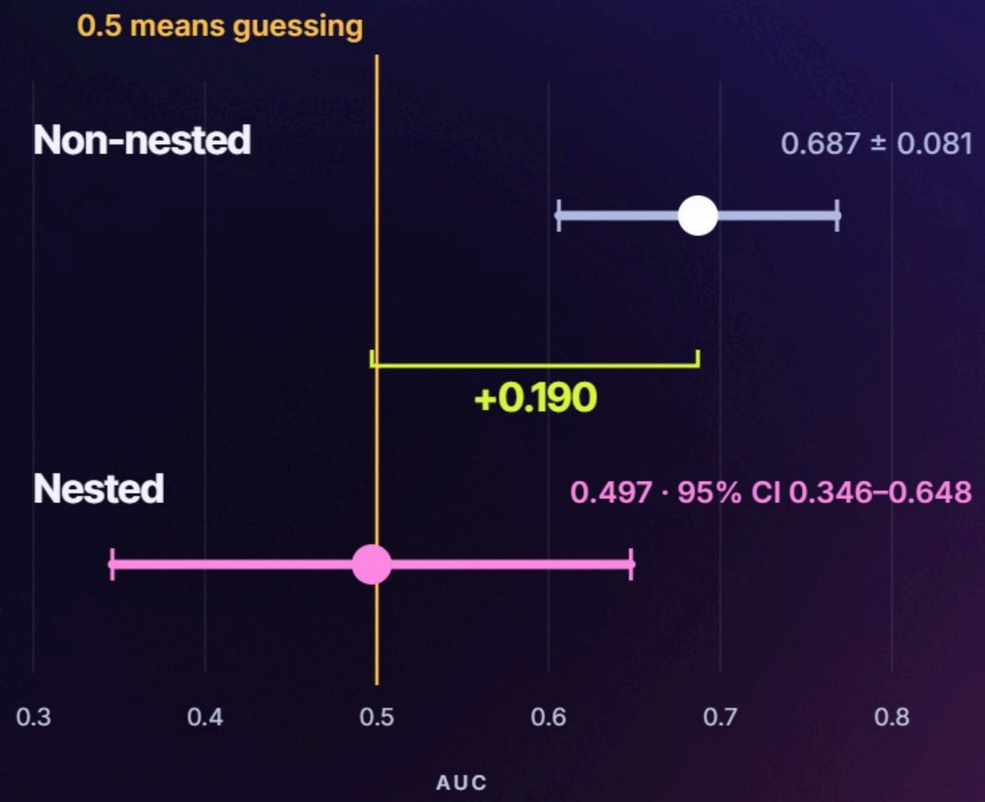
CHECK 02 · LEAKAGE

I used all the data.

The model was not reading the tumour. It was reading my own choices back to me.

Of a hundred patients, about fifty go into the wrong group.

+0.190 — all of it came from the mistake





PROVENANCE

Same images. Different software. Opposite answers.

This fits a software effect, but it does not prove one.

Not 479 independent findings.

Of six genes, KDR was the strongest: q = 0.010

DISCUSSION

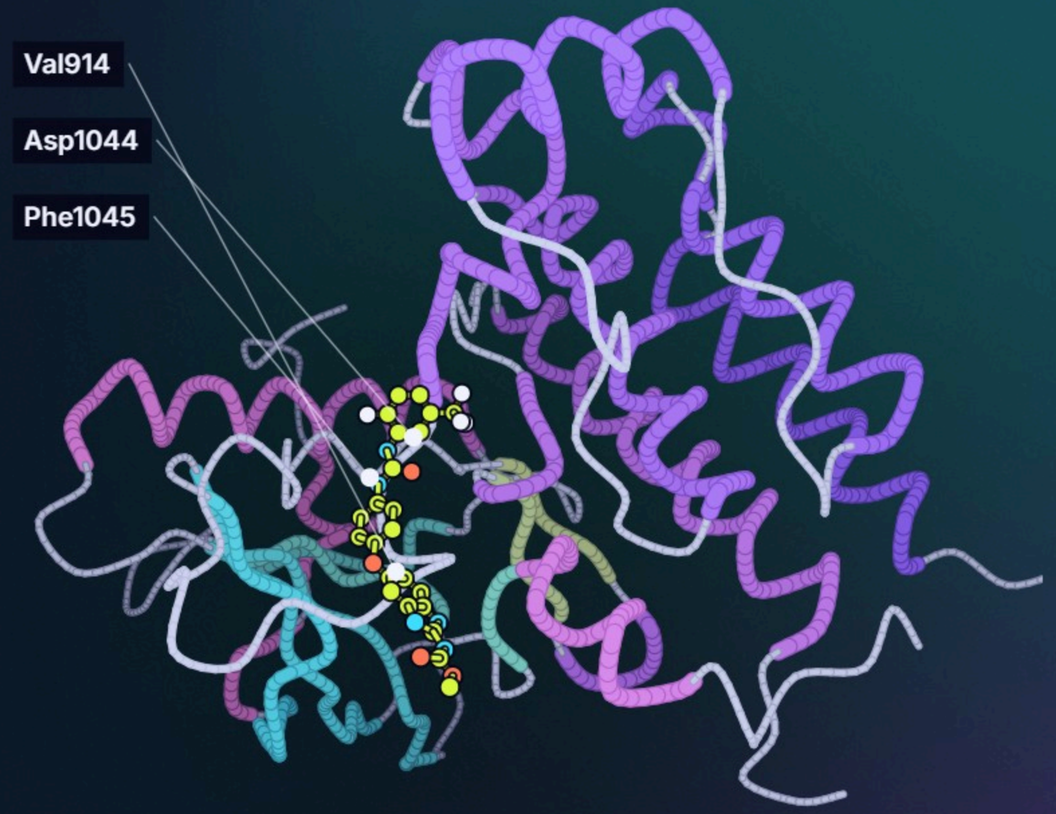
The biology did not change. The record keeping did.

Two ordinary choices, how the measurements were made and how the model was tested, changed the answer completely.

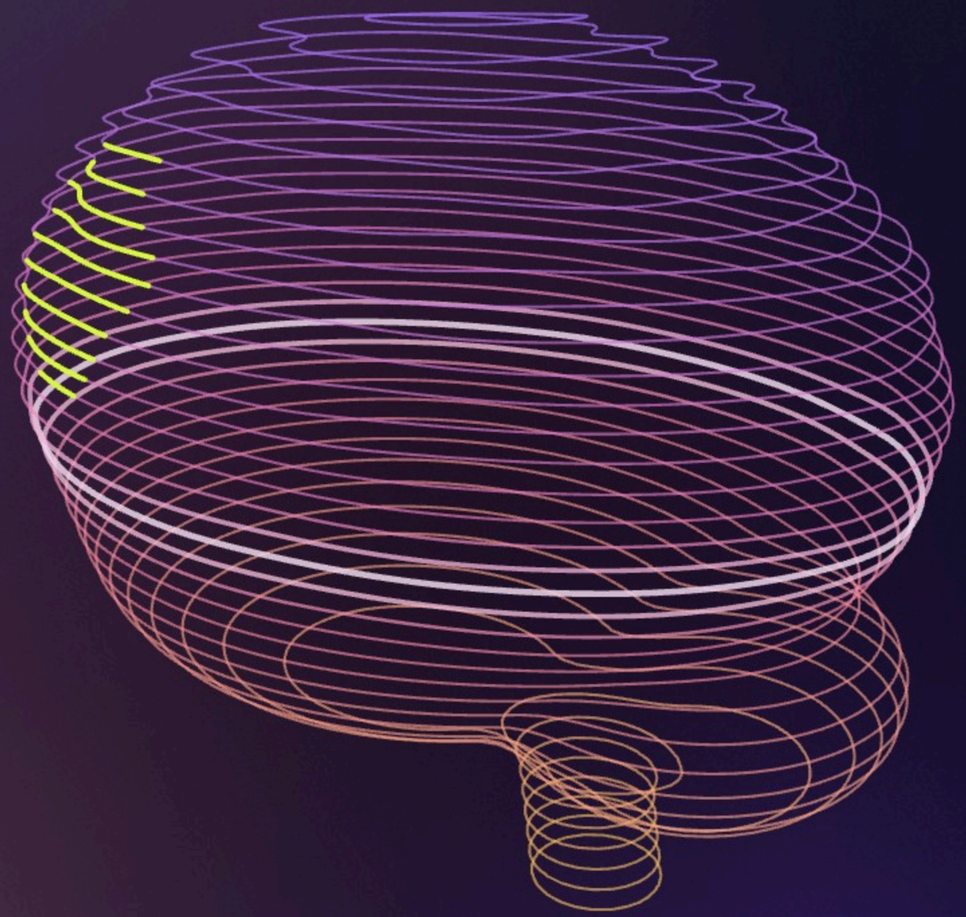
Run both checks before you publish your number.

Mine took one afternoon.

Everything here is public, the code is open. I would be glad if someone tried to prove me wrong.
ipb.link/triu-nugraha-2026



Inhibitor · Val914 · Asp1044 · Phe1045 — VEGFR-2 kinase domain, PDB 2OH4



Thank you.

ขอบคุณครับ

Muhammad Dandy Nugraha · Wisnu Ananta Kusuma
Department of Bioinformatics, IPB University

muhammad_dandy@apps.ipb.ac.id
ipb.link/triu-nugraha-2026

Extra slides: validation design · software comparison · protein structure



BACKUP A · VALIDATION DESIGN

Validation design

Non-nested

Screening and tuning fitted on the full cohort.

AUC 0.687 ± 0.081

Nested

Both confined to each outer training fold.

AUC 0.497 · 95% CI 0.346–0.648 · accuracy 48.4%

Median split z = −0.179 · 46 HIGH / 45 LOW



BACKUP B · SOFTWARE COMPARISON

Software comparison

Table 1. Correlation with KDR expression for two descriptor sets computed from the same images.

DESCRIPTOR SET	PATIENTS	TESTED	NOMINAL	FDR Q<0.05	MIN Q
Pre-computed BraTS	**91**	**704**	**61**	**0**	**0.530**
Re-extracted PyRadiomics	**86**	**1098**	**682**	**479**	**0.010**

479 represents a few underlying dimensions expressed repeatedly, not 479 independent findings.

CXCL8 also survived correction (q = 0.039).



Inhibitor · Val914 · Asp1044 · Phe1045 — VEGFR-2 kinase domain, PDB 2OH4

BACKUP C · PROTEIN STRUCTURE

VEGFR-2 kinase domain, PDB 2OH4.

01 Eight ATP-binding residues within 5 Å of the inhibitor.

02 Gatekeeper Val914

03 DFG motif Asp1044/Phe1045

04 Numbering offset by two from UniProt.

Qualitative visualisation supplying molecular context rather than a mechanistic bridge to the imaging model.